

Operation of conventional and unconventional energy sources to drive a reverse osmosis desalination plant in Sinai Peninsula, Egypt

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ARTICLE INFO

Article history:

Received 2 February 2019

Received in revised form

27 May 2019

Accepted 31 May 2019

Available online 3 June 2019

Keywords:

Water desalination
Hybrid power system
Renewable energy
Reverse osmosis
Homer Pro software

ABSTRACT

Desalination process is an essential demand to overcome the lack of drinking water in remote areas in Egypt. Hybrid energy system drives desalination techniques to provide potable water for islands and coastal zones where there is no electrical grid. This paper analyzes the technical configuration, sizing and economic optimization of off-grid hybrid energy system powered reverse osmosis plant. The hybrid system consists of wind turbines, photovoltaic panels, converters, storage batteries, and a diesel generator. The objective of this study is to satisfy the electrical energy demand of the reverse osmosis (RO) plant with a freshwater capacity of 100 m³/day for use at Nakhl, North Sinai, Egypt. This simulation studied eleven different configurations of energy sources to select the optimal case at the proposed location. Homer package is used to select the cost-effective, sustainable and socially accepted system. The simulation is based on calculating the net present cost (NPC), excess system electricity, cost of energy (COE) and carbon dioxide emissions (CO₂). Through the study, the optimal configuration consists of 160 kW photo-voltaic (PV) panels, 19 strings of lead-acid batteries and a 50 kW diesel generator system. The optimal case achieves the lowest cost of energy (\$0.107/kWh) and net present cost (\$502,662).

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1. Introduction

Oceans are an essential and permanent source of water on our planet. They cover the vast majority of the Earth surface. Their salinity makes a problem for human consumption, agriculture activities and growth of industrialization which require a large amount of freshwater [1]. Nearly to half of the world population will suffer from potable water lack in 2030 [2]. Therefore, researchers and decision makers start looking for desalination techniques as the best way for facing water shortage especially in coastal zones and remote areas [3–5]. Water desalination needs large quantities of energy with smart ways to achieve economic and sustainable operation. Therefore, the hybrid renewable energy system powered desalination processes granted the right way to overcome the freshwater shortage [6].

Desalination techniques are classified as thermal (thermal vapor compression TVC, multi-stage flash distillation MSF and multi-

effect distillation), chemical (ion exchange IE), electrical (electrodialysis ED) and mechanical (mechanical vapor compression MVC and reverse osmosis (RO)). Reverse osmosis uses the pressure gradient to separate the salt particles and produce freshwater [7]. Electro-dialysis and reverse osmosis are the most widespread technologies to be able to overcome the old techniques disadvantages such as high capital cost, adverse effect on aquatic life, environmental pollution and high energy consumption [1]. Natural vacuum desalination, freezing and adsorption desalination and forward osmosis (FO) are new desalination processes [8].

Chaouki Ghenai et al. designed hybrid energy system with the low cost of energy, high renewable fraction and low carbon dioxide gas emissions to serve the load in Sharjah, United Arab Emirates. They concluded that the grid-tied PV system gives the best performance with 47.3% renewable fraction, 90 \$/MWh energy cost and low carbon dioxide emissions (264.25 kg CO₂/MWh) [9]. Murat Gokcek studied the operation of seven stand-alone power systems to satisfy the demand power to RO plant in Bozcaada Islands, Turkey [7]. The results showed that the optimal hybrid power system consists of (20 kW PV panels, one wind turbine with 10 kW rated, an 8.90 kW diesel generator, a 120 kWh lead-acid battery,

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and a 7.35 kW converter).

Mohamed Ali Abdelkareem et al. [1] conducted a study to review the recent progress in water desalination powered by a renewable energy system. A. A. Hossam Eldin [6] studied the technical and economic optimization of a stand-alone hybrid system equipped RO seawater desalination plant in Matrouh city, Egypt. In this study, the lowest cost of energy was found to be 0.282 \$/kWh for a hybrid system consists of PV/wind/diesel with battery and converter. R. Nagaraj et al. [10] studied the design of small desalination plants based on the RO process with renewable energy sources and a backup diesel generator system. The results showed that the best option is a hybrid power system consisting of a

combination of 1 kW wind turbine, 8 kW PV panels and 1.5 kW storage battery with the cost of energy (INR 12.73). Roberta Fornarelli et al. [11] conducted a study in Australian to identify a renewable energy powered water desalination that is sustainable, low cost of energy and socially accepted. The analysis results showed the off-grid hybrid system reduces the cost of electricity by 47% compared to the grid. Gul Ahmad Ludin et al. [12] studied the importance of hybrid system with a backup diesel generator to feed the load in an off-grid rural village in Afghanistan. The results showed that the hybrid renewable system achieved the balance and security due to the lowest life-cycle cost and cost of energy. Baojia Wu et al. conducted a study on the optimal size of a hybrid



Fig. 1. Flow chart for selecting the optimal configuration of water desalination plant energy using Homer Pro software.

energy system powered reverse osmosis plant to serve a load of a stand-alone region in Iran [13]. The results concluded that PV/diesel system with storage batteries powered reverse osmosis desalination system is sustainable, cost-effective and socially accepted in Iran.

All studies showed that renewable energy powered RO desalination system is more reliable and sustainable than others due to different factors such as low operation cost, low carbon dioxide emissions, clean energy system, use the natural resources which are permanent. However, many factors should be taken into consideration before choosing renewable energy such as feed and product water salinity, feed pressure, plant location, the wind speed and solar radiance in the site, plant size and cost of product water [14]. These factors play a vital role in determining the electrical energy required and the proper sources rate.

The paper's primary contribution is reducing the greenhouse gas emissions according to the Paris Agreement which was adopted on December 12, 2015. This agreement aims to strengthen the global response to the threat of climate change by reducing greenhouse gas emissions, which affect food and water, and holding the increase in the global average temperature to well below 2 °C. This study documents that renewable energy is an effective solution to face fuel energy pollution by reducing greenhouse gases emissions by approximately 94%. Besides, this paper presents a completed real case of desalination study at Nakhl which is located in the middle of Sinai Peninsula and is bordered by mountains of all directions. As a result of its geographical location, the city lacks both natural water resources (rivers) and electricity grid connections. So, the city inhabitants rely on the water from wells. Eleven possible scenarios are investigated with and without renewable energy sources and storage systems to get the optimal hybrid system configuration and in the same time to overcome the drawbacks of the majority of the previous studies that concentrate on certain specific few possible scenarios. The study documents that the hybrid energy sources are cheaper than conventional energy sources. Also, the manuscript inspects the economic feasibility of the proposed control scheme in addition to its ability to reduce CO₂ emissions to address the gap of the majority of the previous studies in the literature review that tried to deal with one only of these points.

Homer Pro software package was used to analyze eleven different cases to reach the optimum power system configuration. The rest of the paper is organized as follows: section 2 was divided into three parts. The first part presents the description of input parameters such as site description, load profile, solar radiation, wind speed and diesel system. The second part illustrates the system components, description and specifications such as PV array sizing, wind turbine production, battery bank and fuel consumption of diesel generator. The third part shows the economic analysis such as net present cost, annual discount rate and levelized cost of energy. Section 3 describes the paper results and discussion. Finally, section 4 presents the work conclusion.

2. Methodology

The required energy, for the brackish water desalination plant proposed in this study, is simulated by Homer Pro package. Fig. 1 illustrates the Homer simulation steps to get the optimal power system configuration. This section discusses the input parameter descriptions, the system specifications, and the economic analysis.

2.1. Description of input parameters

Homer Pro software package is used to calculate the optimum results for different systems configurations by simulating the input

parameters. These parameters are summarized in location, electric load demand, solar radiation, wind speed, and diesel.

2.1.1. Site description

The proposed RO desalination plant is assumed to be installed at Nakhl city, Egypt at an elevation of 420.6 m or 1,380 ft (29° 59.4' N, 34° 17.1' E). Nakhl city is one of six towns which are located in North Sinai governorate. It is located along the border of South Sinai governorate with North Sinai governorate in the heart of Sinai Peninsula. Fig. 2 shows the proposed location in this paper from Homer Pro window. El Tih Mountains are surrounding the city from different directions. The town's climate is classified as borderline hot desert and cold desert climate. The site topography is characterized as foothills and mountains with large solar irradiance and wind speed.

2.1.2. Load profile

The daily load of the reverse osmosis desalination plant is built. It consists of the combination of four parts as filter pump, reverse osmosis unit, product water pump and lighting loads. The first stage, pretreatment process, uses a filter pump with 7.64 kW for 16 h per day to prepare desalinated water with permissible percentage reduction of dissolved solids. This pump consumes 122.24 kWh/day. The second stage, the reverse osmosis unit, has two high-pressure pumps with 11.19 kW for each one. This unit works 16 h per day to separate potable water from brackish water. This stage requires 358.08 kWh/day. Finally, product water pump with 3 kW is used for 5 h per day to get the fresh water to Nakhl city. Besides, lighting loads consume 7.65 kW along the day. A sample of the daily load profile for the reverse osmosis plant is presented in Fig. 3.

From the load profile, RO plant consumes low power rate from 12 a.m. to 6 a.m. daily. The maximum load begins at 7 a.m. till 11 p.m. The plant requires a maximum of 35.67 kW peak load with



Fig. 2. The proposed location map (Nakhl, North Sinai Governorate, Egypt) from Homer Pro window.

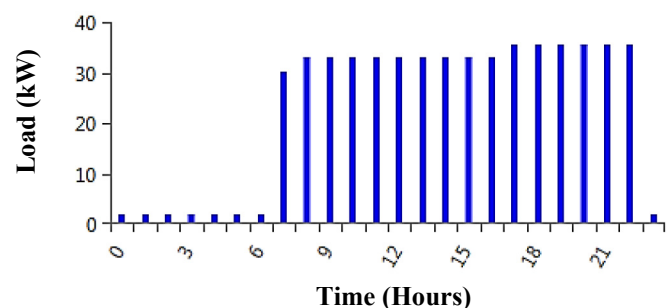


Fig. 3. A sample of the daily load profile for the proposed desalination plant.

average baseline 23.22 kW. Load factor is 0.65 which illustrates the ratio of average load to peak load. The Scaled Annual Average of the plant is 557.22 kWh/day that is equivalent to 203.385 MWh/year.

This system aims to produce 100 m³/day of fresh water to meet the requirements of life at Nakhl city. In Semi-dry conditions, the world health organization (WHO) and united nations (UN) recommend that the adult man requires about 20 L of freshwater a day. Therefore, 5,000 people can benefit from this study [15].

2.1.3. Solar radiation

National aeronautics and space administrative (NASA) is a trusted source for renewable energy parameters [16]. It presents the required data to evaluate the energy produced by renewable energy sources such as photovoltaic panels and wind turbines.

The clearness index is considered an essential factor to measure the atmosphere clearness. It ranges between zero and one [17]. Fig. 4 shows the monthly average solar radiation and clearness index data which are fed to Homer Pro software. At Nakhl site, the solar radiation ranges from 6.12 kWh/m²/day at January to 9.81 kWh/m²/day at September with annual average value equals 7.81 kWh/m²/day. According to Fig. 4, the average clearness index is 0.66. The solar irradiance in the proposed region is high from March to October, while it has low solar radiation from November to January. Monsoon season controls the solar radiation values [18].

2.1.4. Wind speed

The wind turbine is a new trend in producing electrical energy to face fossil fuel pollution, high energy costs, and energy crisis [19]. Wind energy serves 0.4% of total global energy demand. It meets the energy required for 35 million people all over the world [20]. Since 2001, Egypt making significant efforts to get benefit from the wind energy. Researchers and policymakers are planning to increase wind energy production to 12% by 2022.

The wind speed data at Nakhl city are obtained from the NASA database [16]. According to NASA measurements, the wind speed varies between 12.78 m/s in January and 2.27 m/s in November at 50 m above the surface of the Earth. Fig. 5 shows that the monthly average wind speed is 5.07 m/s at the proposed site. Fig. 6 presents

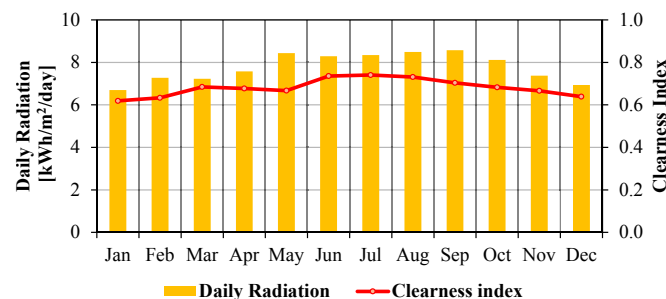


Fig. 4. The monthly average solar radiation and the clearness index data from NASA at Nakhl, Egypt.

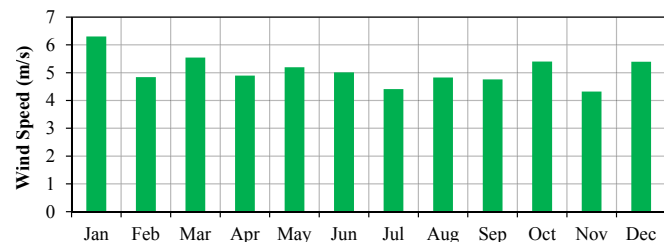


Fig. 5. The monthly average wind speed data from NASA at Nakhl, Egypt.

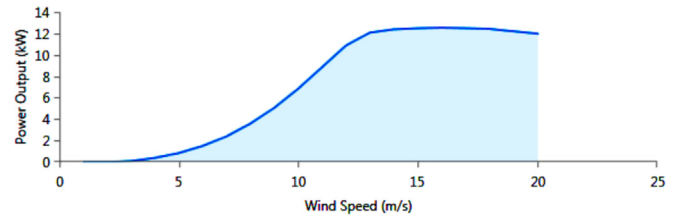


Fig. 6. The power-speed curve of Bergey excel (XL10) wind turbine.

the output power curve of the wind turbines generator.

2.1.5. Diesel

Due to the economic reform in Egypt, the Egyptian Government reduces the financial supports on the petroleum products. The diesel price in Egypt is L.E. 5.5/L or \$0.31/L at the year 2018 [21].

2.2. System components, description, and specification

The proposed system consists of a hybrid energy system powered reverse osmosis plant. The hybrid system includes solar panels, wind turbines, backup diesel generator and a conditioning unit (converter). Although the battery banks increase the total cost of the stand-alone (off-grid) system, they are used to reduce the gas emissions and fuel consumption by reducing the diesel generator start/stop cycles [22,23].

Reverse osmosis plant consists of a raw water reservoir, pre-treatment unit, high-pressure pump (HP), RO membrane unit and storage tank. Table 1 shows the technical specifications of the RO desalination system [24,25]. The component descriptions, technical specifications, and parameter costs of the proposed hybrid system are listed in Table 2 [26–31].

2.2.1. PV array sizing

Egypt enjoys excellent solar radiation along the year which encourages investing the renewable energy to convert solar radiation into electricity. Therefore, the electrical output power of photovoltaic arrays can be calculated as follows:

$$P_{\text{output}} = P_{\text{rated,STC}} f \left(\frac{G_i}{G_{i\text{-STC}}} \right) [1 + \alpha_p (T_c - T_{c,\text{STC}})] \quad (1)$$

where $P_{\text{rated,STC}}$ is the rated output capacity (kW) under standard test condition (STC), f (%) is the derating factor of PV (compensate the reduction in PV module efficiency), G_i is the incident solar radiation on the required module (w/m²), $G_{i\text{-STC}}$ is the total

Table 1

Technical specifications of the RO desalination system.

Characteristic	Unit	Value
Feed pressure	Bar	21
Pump efficiency	–	0.85
Feed flow	m ³ /h	10
Permeate flow	m ³ /h	100
Recovery	%	65
Power requirement	kW	32.84
Energy Consumption	kWh/m ³	507.32
Membrane manufacture	–	FILMTEC.-BW30-400
Membrane element numbers	–	6
Feed temperature	°C	27–45
Feed pump consumption	kW	7.64
High-pressure Consumption	kW	22.38
Product water power consumption	kW	3
Auxiliary loads	kW	7.65
Daily operating hours	hours	16

Table 2
Technical and economic specifications of the proposed power system.

Description	Unit	Specification
PV modules		
PV model	—	Peimar SG325P
Rated power	kW	1
Capital cost	\$/kW	550
Replacement cost	\$/kW	550
Operating and maintenance cost	\$/year	10
Lifetime	years	30
Derating factor	%	80
Ground reflection	%	20
Efficiency	%	16.7
Wind turbine		
Type of wind turbine	—	Bergey Excel 10
Rated power	kW	10
Capital cost	\$/unit	67860
Replacement cost	\$/unit	67860
Operating and maintenance cost	\$/year	7000
Lifetime	year	20
Diesel generator		
Generator model	—	Generic fixed capacity
Rated power	kW	50
Capital cost	\$/kW	500
Replacement cost	\$/kW	500
Operating and maintenance cost	\$/hour	0.03
Lifetime	hours	15000
Minimum load ratio	%	25
Diesel fuel price	\$/L	0.31
Storage battery		
Type of battery	—	Discover 12VRE-3000TF
Capital cost	\$/unit	410
Replacement cost	\$/unit	410
Operating and maintenance cost	\$/year	10
Lifetime	years	20
Converter		
Converter model	—	Generic converter
Capital cost	\$/kW	300
Replacement cost	\$/kW	300
Operating and maintenance cost	\$/year	10
Lifetime	year	15
Conversion efficiency	%	95

irradiance under reference conditions (STC), α_p is the power temperature coefficient ($^{\circ}\text{C}^{-1}$), T_c is the PV module temperature ($^{\circ}\text{C}$) and $T_{c,STC}$ is the PV module temperature under STC [9,17,31]. All data are used by Homer to specify the PV capacity. PV panel is designed with a 29.99° panel slope and no tracking system.

The proposed PV module is 72-cell polycrystalline (PV-SG325P) with 325 W_p nominal output power. The energy system combines with 806 PV modules which connected in series to provide the desalination plant with the required energy. The module area is 1.941 m^2 with a total area of 1,565 m^2 [32].

The energy production from solar cells is negatively affected by the ambient temperature [33,34]. Therefore, the temperature coefficient ($-0.43\%/^{\circ}\text{C}$) is taken into account. Under standard test condition, PV array efficiency is 16.7% with 22.5 kg footprint for 1.951 m^2 . Table 3 shows the technical characteristics of the PV module under standard test condition (STC) (Irradiance $G_{STC} = 1000 \text{ W/m}^2$, Module temperature $T_{c,STC} = 25^{\circ}\text{C}$, Air mass = 1.5)

Table 3
Technical characteristics of the PV-SG325P module.

Nominal output (P_{max})	W	325
Voltage at P_{max} (V_{mp})	V	36.3
Current at P_{max} (I_{mp})	A	8.96
Dimensions	mm	1957*992*40
Weight	kg	22.5
NOCT	$^{\circ}\text{C}$	45 $\pm$ 2
Operating temperature	$^{\circ}\text{C}$	-40 ~ +85

[32].

2.2.2. Wind turbine production

The output power of wind turbine depends on air density, wind speed and hub height [31]. In this study, Bergey excel XL10 (AC type) wind turbine is used. Table 4 illustrates the technical characteristics of the proposed wind turbine [35].

2.2.3. Battery bank

Storage battery meets the system energy requirements at night when no solar radiations for PV panels or no wind to operate the wind turbine. Table 5 presents the technical characteristics of the battery [36,37]. In this paper, Discover 12VRE-3000TF is used as a storage battery. The cells are connected in series with string size equals ten batteries. Each string produces 31.1 kWh to provide the required energy.

2.2.4. Fuel consumption of diesel generator

Energy continuity is an essential factor in establishing a reliable system. So, the diesel generator is used as a backup power source to compensate the capacity shortage due to the absence of solar and/or wind energy [38]. Diesel fuel consumption hourly cost (L/h) depends on the consumption rate. So, it can be calculated by using the following equation [39].

$$C_F = P_f F_{gen} \quad (2)$$

In the last equation, P_f is the fuel price (\$/L), and F_{gen} is the fuel consumption rate which depends on the output power of the generator. It can be expressed by Ref. [38].

$$F_{gen} = f_{gen0} p_g + f_{gen1} p_r \quad (3)$$

where p_g and p_r are the generator output and rated power respectively. Also, f_{gen0} is the intercept coefficient of the fuel curve which equals 0.246 L/kWh, and f_{gen1} is the fuel curve slope which equals 0.08145 L/kWh.

2.2.5. Power conditioning unit

In this simulation, the Generic converter is used for inversion and rectification processes. Inversion process is used with PV arrays to convert the electric current from DC to AC for supplying the load.

Table 4
Technical characteristics of the Bergey Excel (XL10) wind turbine.

Design life	years	20
Rotor diameter	m	7
Tower types	—	Guyed and Non-guyed
Hub height	m	24 ~ 49 (30 m is used for study)
Rated power	kW	10
Orientation	—	Upwind, 3 Blade, Horizontal Axis
Furling wind speed	m/s	14 ~ 20
Cut-in wind speed	m/s	2.5
Cut-out wind speed	m/s	None
Max. design wind speed	m/s	60

Table 5
Technical characteristics of the Discover 12VRE-3000TF battery.

Nominal voltage	V	12
Nominal capacity	kWh	3.11
Maximum capacity	Ah	260
Round trip efficiency	%	80
Maximum charge current	A	43
Lifetime throughput	kWh	3581.60
Lifetime	years	20
Minimum state of charge	%	20

Rectification process is used with a diesel generator and wind turbine to convert the electric current from AC to DC for charging the battery [37]. Homer optimizer selects the proper capacity for the converter in this study.

2.3. Economic analysis

The discount rate, inflation rate and project lifetime are very vital parameters for economic analysis. These parameters are needed for Homer simulation to evaluate the net present and the cost of energy. Homer classifies the power system configurations

according to the total net present cost. Also, the cost of energy plays a vital role in selecting the optimal power system configuration.

2.3.1. Net present cost (NPC)

The net present cost (as Homer definition) is the life cycle cost which is the variance between the present amount of payments during the project age and the present amount of income during the same period. Project payments are capital, operating and maintenance costs. The following formula can obtain the net present cost [37,40]:

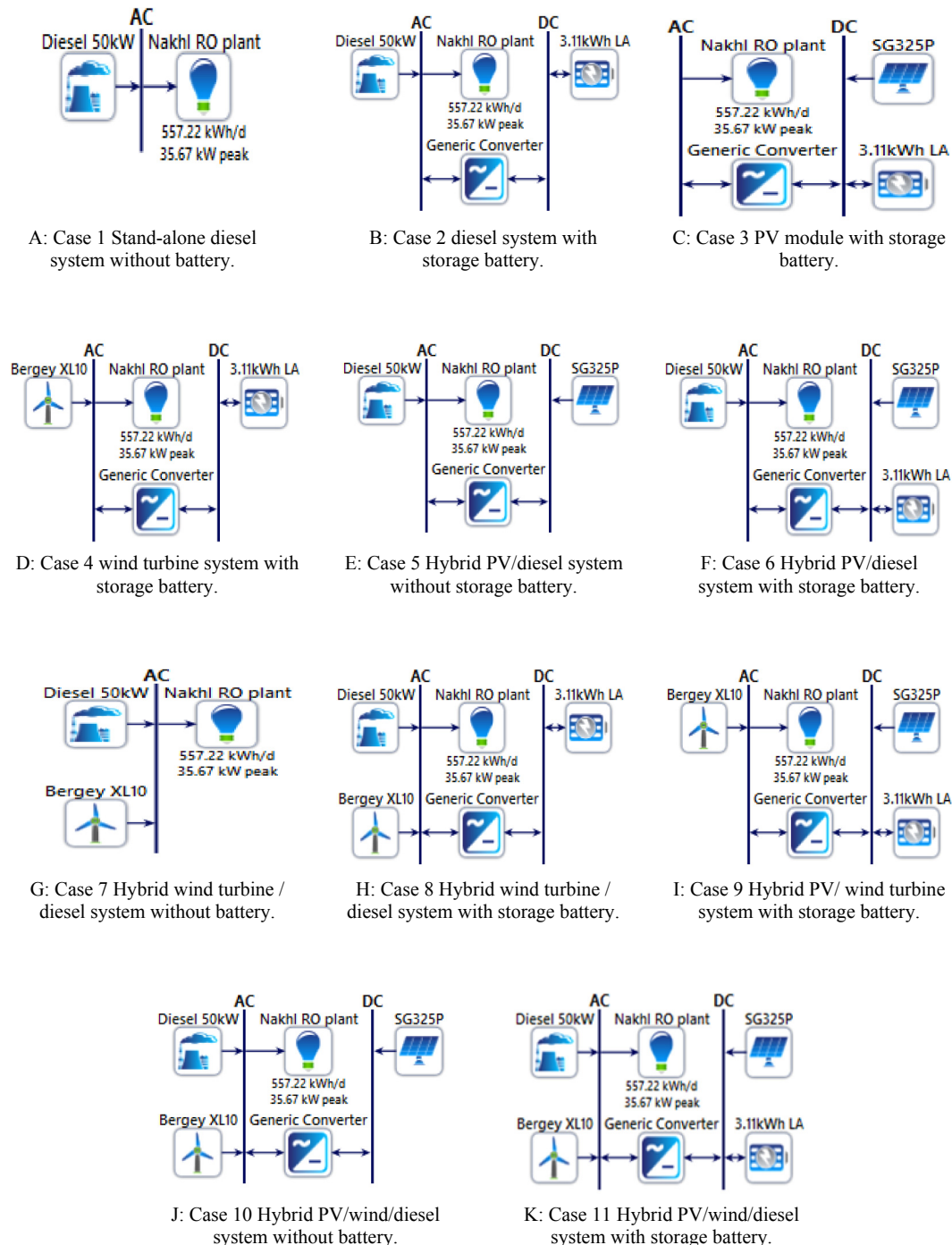


Fig. 7. Different scenarios of the stand-alone hybrid energy system at Nakhl city.

$$C_{NPC} = \frac{C_{ann,tot}}{CRF(i, n)} \quad (4)$$

where $C_{ann,tot}$ is the total annualized cost (\$/year) which consists of all costs during the project lifetime such as capital, annual operating and maintenance, replacement and fuel costs. $CRF(i, n)$ is the capital recovery factor (as Homer definition) which represents a series of equal annual income. It can be estimated using the following formula [41]:

$$CRF(i, n) = \frac{i(1+i)^n}{(1+i)^n - 1} \quad (5)$$

where i is the real discount rate. Also, n is the project lifetime which equals 25 years in this simulation.

2.3.2. Real discount rate (i)

Homer defines the real discount rate as the real interest rate which is used to convert between annualized and in-time costs. Homer calculates the real interest rate according to the following equation [37,40].

$$i = \frac{i' - f}{1 + f} \quad (6)$$

In the above equation, i' is the nominal interest rate and f is the annual inflation rate. In Egypt, the annual inflation rate and the real discount rate as referred in October 2018 are 16% and 16.75% respectively [42].

2.3.3. Levelized cost of energy (COE)

Levelized cost of energy is the ratio between the total annualized cost (\$/kWh) and the total serviced load. The following equation can express the levelized cost of energy [37,40]:

$$COE = \frac{C_{ann,tot}}{Load\ served} \quad (7)$$

where $C_{ann,tot}$ is the total annualized cost (\$/year) and $Load\ served$ is AC, and DC electricity served.

3. Results and discussion

This paper studied different scenarios for supplying the

proposed RO desalination plant with the required energy to produce 100 m³/day of freshwater. The results are obtained by comparing the optimal configuration of stand-alone diesel system with and without storage battery, photovoltaic panels with battery, wind turbines system with storage battery, hybrid diesel/PV system with and without storage battery, hybrid diesel/wind turbine system with and without storage battery, PV/wind system with storage battery and hybrid PV/wind/diesel system with and without storage battery. The capacity of the PV system is changed from 153 kW to 379 kW. The number of wind turbines is varied from one turbine to 32 turbines. The storage system consists of 10 batteries for each string. The string size is varied from one string to 303 strings. A fixed capacity generator is used. Homer auto-size mode is used to select the optimal converter capacity. The project lifetime is 25 years.

3.1. Case 1: stand-alone diesel system without storage element

The stand-alone diesel system is connected to the load through the AC bus bar as shown in Fig. 7A. Although this system has the lowest capital cost (\$25,000), it ranks in 8th position between cases due to its high production of gas emissions. Also, the fuel price has a direct effect on the net present cost (NPC) which equals \$1,200,435. According to Global Petrol Prices, the simulation assumed that the fuel price is \$0.31/L at Nakhil city, North Sinai government, Egypt. The annual energy demand is AC load which requires 203,385.3 kWh/year. In this simulation, the mean electrical output power of the generator is 26.7 kW. The generator is operated for 8760 h/year with one start/year to produce total energy equals 234,045 kWh of electricity/year with 100% percentage. Therefore, the generator covered the demand energy and generated excess electricity equals 30,660 kWh/year (13.1%). So, the percentage of unmet electric load and capacity shortage is zero. The diesel consumed 78,348 L of fuel/year. The carbon dioxide emission (CO₂) is 205,106 kg/year. Unfortunately, the excess electricity is lost due to the battery-less system. Fig. 8 illustrates the cash flow summary of the stand-alone diesel system which is categorized by cost type. The results tell us that the fuel price represents about 46.5% of the total cost. Therefore, NPC of the stand-alone diesel system is higher than the other systems.

3.2. Case 2: diesel system with a storage battery

Fig. 7B shows the configuration of the diesel system with a

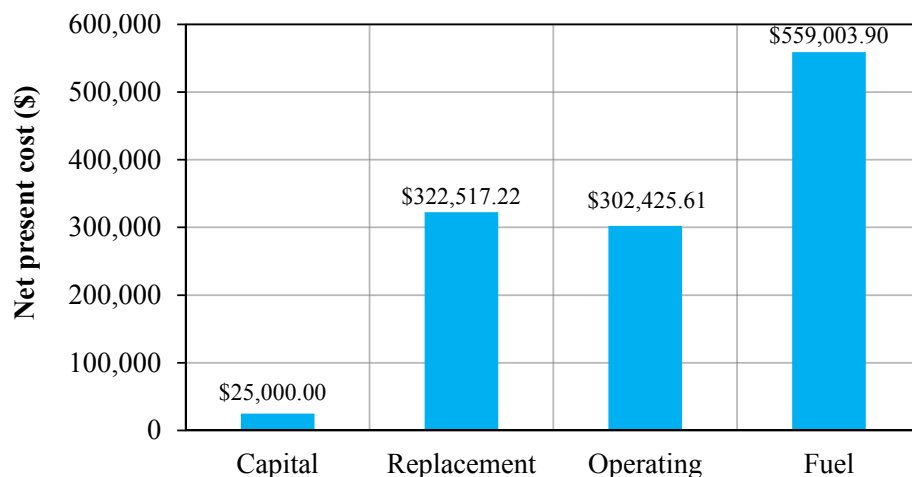


Fig. 8. Cash flow summary of the stand-alone diesel system.

storage battery. This system ranks the 5th position between cases. It produced 100% of energy with 205,628 kWh/year. The generator is operated for 5840 h/year. It consumed 65,773 L of fuel/year with gas emissions (CO₂) equal 172,184 kg/year. All these values are better than the values in the previous case due to the storage battery. A string of 10 batteries is used to meet the load. Fig. 9 shows that the lighting loads (12 a.m.–6 a.m.) are supplied from the battery while the generator served the rest of the loads from 6 a.m. to 11 p.m. The system NPC is \$919,902.20 with a cost of energy (COE) equals \$0.1965. Also, the fuel price represents 51% of the total cost.

3.3. Case 3: PV system with a storage battery

Although the high capital cost of this system, it ranks the 2nd position between cases due to the high renewable fraction percentage (100%). Fig. 7C illustrates the system configuration. It consists of photovoltaic arrays with 262 kW rated capacity, 56 strings of batteries and converter with 39.4 kW rated capacity. The PV panels are operated for 4390 h/year to produce 466,349 kWh of electricity per year. Fig. 10 shows the monthly average electric production of the PV system. It is noted that the peak PV output power occurred from April to September during the summer season. Zero gas emissions characterize this system. Unfortunately, 0.0392% of electric load is not serviced due to the capacity shortage (0.0978%).

3.4. Case 4: wind turbine system with a storage battery

The system configuration is shown in Fig. 7D. This system has the highest net present cost (\$10,167,220) in this study. So, it ranks the last position between cases. It consists of 32 wind turbines, 303 strings of batteries and a converter with 110 kW rated capacity. The wind turbines are operated for 7,314 h/year to provide 261,661 kWh of electricity per year. Although there are zero gas emissions and 100% renewable fraction, there are 0.0792% of unmet electric load due to the capacity shortage (0.0975%). Fig. 11 shows that the peak



Fig. 11. Wind turbine (XL10) monthly average electric production.

output power from the wind turbine occurred in May while the minimum output power is delivered in December.

3.5. Case 5: hybrid PV/diesel system without storage element

Fig. 7E shows the hybrid system configuration. This system ranks the 6th position between cases. The diesel is operated for 6714 h/year to produce 131,383 kWh of electricity per year with 16.3% percentage. It consumed 46,946 L of fuel with 645 starts/year. The carbon dioxide emission is 122,897 kg/year. Also, PV panels output power is 379 kW. They are operated for 4,390 h/year to produce 676,056 kWh of electricity per year with 83.7% percentage. Although the system renewable fraction is 35.4%, the excess electricity is lost due to the battery-less system. This system should be modified to overcome the stability problems due to the high renewable penetration.

3.6. Case 6: hybrid PV/diesel system with a storage battery

The system configuration is shown in Fig. 7F. It ranks the 1st position between cases. It has the lowest net present cost (NPC) (\$502,661.50) and the cost of energy (COE) (\$0.1074). This case is characterized by a high renewable fraction (93.1%). Fig. 12 shows the monthly average electric production of the system. The diesel is operated for 519 h/year with mean electrical output power equals 27.2 kW to produce 14,109 kWh of electricity per year with 4.7% percentage. Fig. 13 shows the diesel output power for a certain period for the optimal case. The PV system is operated for 4390 h/year with mean output power equals 32.7 kW to produce 286,119 kWh of electricity per year with 95.3% percentage. Fig. 14 illustrates the output power of the PV system for a certain period for the optimal case. The battery system is used to store the excess electricity to maintain the system stability constant and supply energy to the load in critical conditions such as faults or capacity

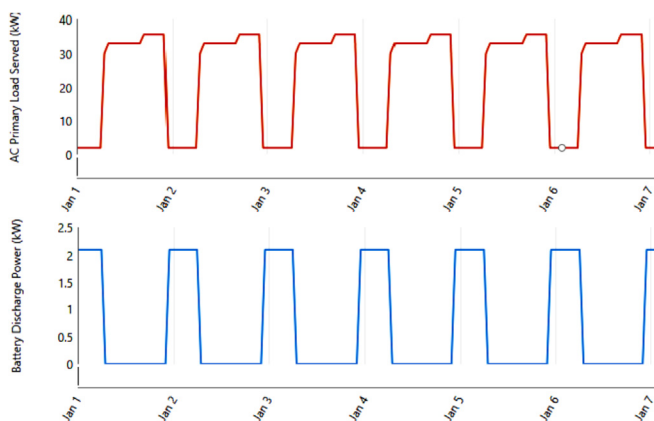


Fig. 9. The battery discharge power with the AC primary served load.



Fig. 10. PV (SG325P) monthly average electric production.

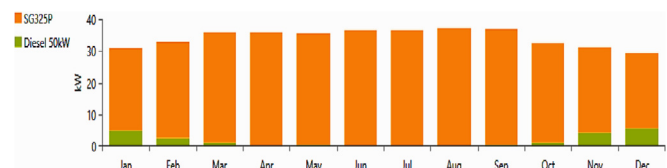


Fig. 12. Monthly average electric production of the optimal system.

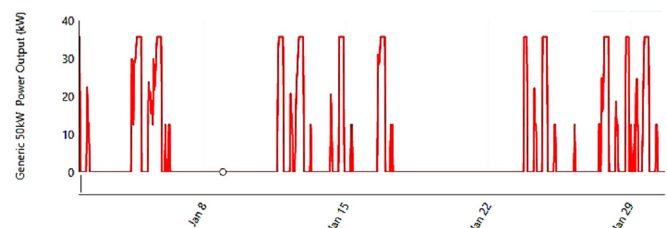


Fig. 13. The diesel output power participation in the optimal case.

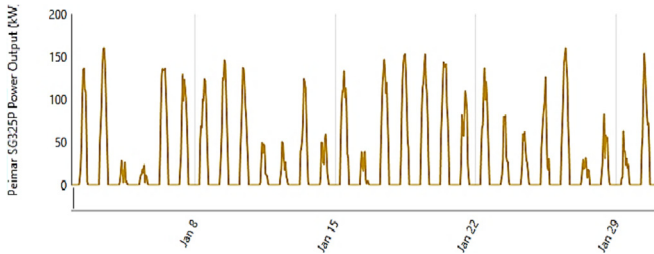


Fig. 14. The PV output power participation in the optimal case.



Fig. 15. The batteries state of charge in the optimal case.

shortage. Fig. 15 shows that the batteries are charged through the year due to the activity of the PV system.

3.7. Case 7: hybrid wind turbine/diesel system without storage element

The wind turbine and diesel system are connected to the load through AC bus bar as shown in Fig. 7G. This system ranks the 10th position between cases. It has a renewable penetration high enough to cause stability problems due to the absence of a storage element. Adding some form of storage devices, such as a battery bank or flywheel, may help.

3.8. Case 8: hybrid wind turbine/diesel system with a storage battery

The system configuration is shown in Fig. 7H. It ranks the 7th position between cases. The diesel system is operated for 5840 h/year with 33.8 kW mean output power to produce 197,173 kWh/year with 96% percentage. It consumed 63,464 L of fuel with 365 starts/year. Also, a wind turbine with 0.933 kW means output power is used to produce 8177 kWh of electricity/year with 3.98% percentage. Fig. 16 shows the monthly average power output from

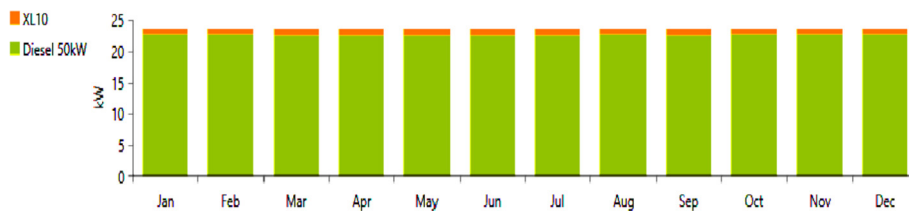


Fig. 16. Monthly average electric production of hybrid wind turbine/diesel system with storage batteries.

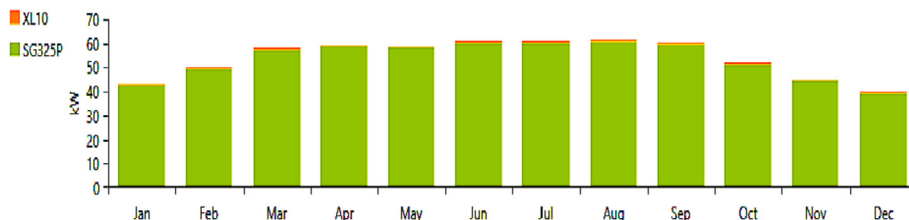


Fig. 17. Monthly average electric production of hybrid PV/wind turbine system with storage batteries.

the generator and wind turbine respectively. This system has a low renewable fraction (3.10%) and excess electricity (0.0995%). The reduction of carbon dioxide emission is increased to 16.9% by comparing with case 7. Although this case has a renewable energy source, its results are very close to case 1 (without renewable energy source). So, this model should be enhanced to be reliable and stable.

3.9. Case 9: hybrid PV/wind system with a storage battery

The system configuration can be simulated as shown in Fig. 7I. This system ranks the 4th position between cases due to no gas emissions. The PV system produced 466,349 kWh/year with 98.3% percent while the wind system produced 8,177 kWh of electricity per year with 1.72% percentage. The renewable fraction is 100% with excess electricity equals 240,193 kWh/year (50.6%). Fig. 17 shows the participation of each element in the system to produce the required energy. It is noted that the photovoltaic system is more reliable than the wind turbine system. A large number of batteries (540 batteries) is used which leads to increase the capital cost to \$445,100.63. Therefore, the system net present cost is increased to \$856,573.10. It means that much money is spent on storage elements only with 46.57% of the total cost.

3.10. Case10: hybrid PV/wind/diesel system without storage battery

The system is modeled as shown in Fig. 7J. It ranks the 9th position between cases due to its net present cost and gases emissions. The hybrid system produced 801,618 kWh/year from different components. There is excess electricity with 74% without storage element. So, this system will face stability problems that require more detailed modeling such as storage elements.

3.11. Case11: hybrid PV/wind/diesel system with a storage battery

This system is simulated as shown in Fig. 7K. It ranks the 3rd position between cases due to its low net present cost (\$733,217.40) and high renewable fraction (92.5%). The diesel is operated for 583 h/year with 26.1 kW mean output power to produce 15,192 kWh of electricity per year with 5.11% percentage. It consumed 5,109 L of fuel with 150 starts/year. The carbon dioxide emissions are 13,376 kg/year. Also, The PV system is operated for

4390 h/year to produce 273,651 kWh of electricity per year with 92.1% percentage. The excess electricity (22%) is stored in the storage batteries to improve the system reliability. Fig. 18 illustrates that the maximum battery chargers rate is started from April to August due to the activity of the PV system.

3.12. Optimal case analysis

Fig. 19 illustrates the reduction of carbon dioxide emissions in the optimal case (case 6) over the other cases. Fig. 20 shows that case 6 has the lowest net present cost in this study. Also, Case 6 has

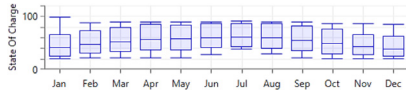


Fig. 18. The batteries state of charge throughout the year.

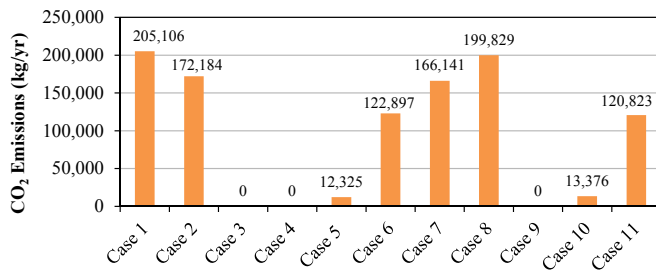


Fig. 19. The value of carbon dioxide emissions in each case.

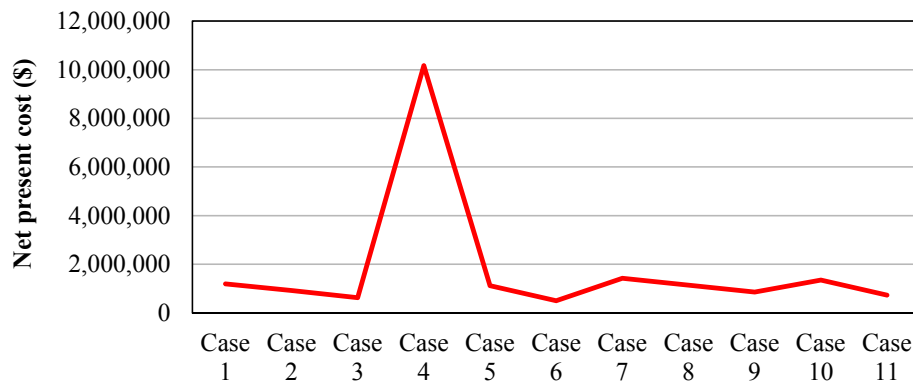


Fig. 20. The net present cost of all cases in this study.

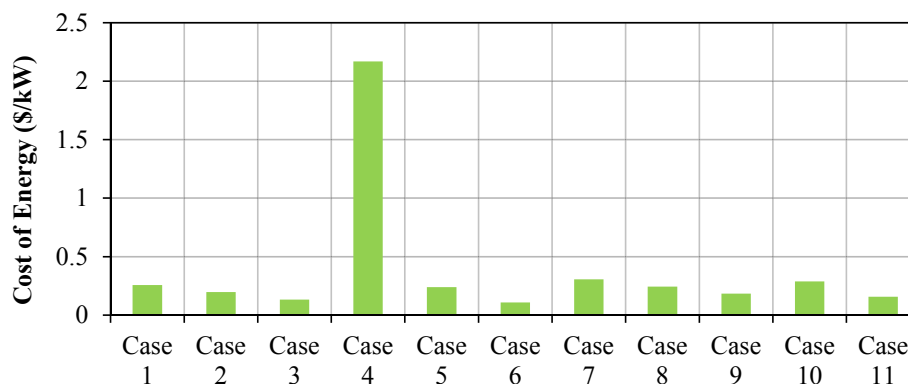


Fig. 21. The cost of energy of all cases in this study.

the lowest COE as shown in Fig. 21. Table 6 illustrates the summary of the optimization results from Homer Pro software. The optimal case (case 6) is bolded in Table 6 to be compared with other cases.

The system economic analysis (NPC, COE, OC) is performed by using the following equations:

$$C_{NPC} = \frac{C_{ann,tot}}{CRF(i,n)} \quad (8)$$

where $C_{ann,tot}$ is the total annualized cost (\$/year) and $CRF(i,n)$ is the capital recovery factor.

$$CRF(i,n) = \frac{i(1+i)^n}{(1+i)^n - 1} \quad (9)$$

where i is the real discount rate and n is the project lifetime.

$$i = \frac{i' - f}{1 + f} \quad (10)$$

where i' is the nominal interest rate and f is the annual inflation rate.

$$COE = \frac{C_{ann,tot}}{Load\ served} \quad (11)$$

where $C_{ann,tot}$ is the total annualized cost (\$/year) and $Load\ served$ is AC and DC electricity served.

4. Conclusion

Water desalination plays a vital role in solving the lack of fresh

Table 6
Summary of the optimization results.

Case	Architecture					Cost				System	
	SG325P (kW)	XL10	Diesel 50 kW (kW)	3.11 kWh LA	Generic converter (kW)	COE (US\$)	NPC (US\$)	OC (US\$/yr)	Initial capital (US\$)	RF (%)	Total fuel (L/yr)
1	—	—	50	—	—	\$0.256	\$1.20M	\$51,071	\$25,000	0	78,348
2	—	—	50	10	2.20	\$0.197	\$919,902	\$38,675	\$29,761	0	65,773
3	262	—	—	560	39.4	\$0.134	\$625,645	\$10,444	\$385,272	100	0
4	—	32	—	3,030	110	\$2.17	\$10.2M	\$291,993	\$3.45M	100	0
5	379	—	50	—	36.7	\$0.239	\$1.12M	\$38,018	\$244,553	35.4	46,946
6	160	—	50	190	39.3	\$0.107	\$502,662	\$13,022	\$202,941	93.1	4,708
7	—	1	50	—	—	\$0.305	\$1.43M	\$58,026	\$92,860	0	76,333
8	—	1	50	10	2.22	\$0.245	\$1.14M	\$45,500	\$97,625	3.05	63,464
9	262	1	—	540	39.9	\$0.183	\$856,573	\$17,878	\$445,101	100	0
10	373	1	50	—	36.6	\$0.289	\$1.35M	\$45,279	\$308,879	36.7	46,153
11	153	1	50	170	39.8	\$0.157	\$733,217	\$20,608	\$258,913	92.5	5,109

Operating cost (OC), renewable fraction (RF).

water in coastal and remote areas. A large amount of energy is essential to produce the necessary potable water. Reverse osmosis desalination plant powered by hybrid renewable energy is proposed to produce 100 m³/day of potable water at Nakhl city, Egypt. In this paper, different energy systems (diesel, PV, wind turbine and battery) are used to select the optimum configuration and economic sizing of the RO plant. The studied systems included a stand-alone diesel system with and without storage battery, photovoltaic panels system with battery, wind turbines system with storage battery, hybrid diesel/PV system with and without storage battery, hybrid diesel/wind system with and without storage battery, PV/wind system with storage battery and hybrid PV/wind/diesel system with and without storage battery. The study is based on the analysis of three main factors such as economic, technical and environmental trends. Case 6 achieved these factors successfully. Also, the decision makers in Egypt are expanding the use of renewable energy sources to compensate for the deficit in petroleum products. The next step is the practical implementation in the same place of the case study thanks to the Egyptian government projects there. Homer Pro software simulated more than 300 solutions. Some of them were feasible, and the others are omitted due to different factors such as lack of converter, capacity shortage constraints, minimum battery life and no sources of power generation.

After the accurate simulation, the following points can be illustrated:

- 1 The hybrid of PV/diesel/battery system in case 6 is the optimal configuration for RO desalination plant at Nakhl city, Egypt.
- 2 Case 6 consists of 160 kW PV panels, a 50 kW fixed capacity diesel generator, 19 strings with 190 lead-acid batteries (3.11 kWh rated capacity) and a 39.3 kW converter.
- 3 This system reduces the emission of carbon dioxide (CO₂) gases of approximately 94%.
- 4 The renewable fraction (RF) on this system is 93.1% which helps to overcome the absence of the electricity grid in remote and arid areas.
- 5 The optimal system meets 100% of the demand energy with zero capacity shortage.
- 6 The Excess electricity in the optimal system can be stored in batteries bank to face the urgent problems. This electricity improved the system stability and reliability.
- 7 The optimal system has the lowest net present cost (NPC) and cost of energy (COE) which equal \$502,661.50 and 0.1074 \$/kW.

Acknowledgment

The authors thankfully acknowledge the support and

cooperation of the University of Ryukyus and the Electrical Power Engineering Department of Zagazig University for their great help.

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